

DAFFODIL INTERNATIONAL UNIVERSITY



LAB REPORT

DEPARTMENT OF COMPUTING AND INFORMATION SYSTEM



LAB REPORT- 02

Course Title : Structured Programming

Course Code : CIS 115

Lab Report : Math.h , If ,If-else , Conditional Operator

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Section: 23A

Semester: 1st, FALL 2025

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Submission Date: 20/10/2025

Submission time: 5:30 pm

Math.h

Problem No: 3.1

Problem Name: Write a program that read any integer number and display absolute value.

Source Code:

```
#include<stdio.h>

int main() {
    int n;
    printf("Enter a number :");
    scanf ("%d", &n) ;
    if (n<0) {
        n=n* (-1) ;
    }
    printf("The absolute value is %d",n) ;
    return 0;
}
```

Output:

```
PS H:\Programming\Coding Practice\Basic C> cd "h:\Programming\Coding Practice\Basic C\exerise class work\3\" ; if ($?) { gcc code1.c -o code1 } ; if ($?) { .\code1 }
Enter a number :-9
The absolute value is 9
PS H:\Programming\Coding Practice\Basic C\exerise class work\3> █
```



Problem No: 3.2

Problem Name: Write a program that read any angle t and display sin(t).

Source Code:

```
#include<stdio.h>
#include<math.h>
int main() {
    double t,pi=3.14159265358979323 ,radians;
    printf("Enter any angle in degree:");
    scanf("%lf",&t);
    radians = t * pi /180.0;
    double result=sin(radians);
    printf("the sin value is: %lf",result);
    return 0;
}
```

Output:

```
PS H:\Programming\Coding Practice\Basic C\exerise class work\3> cd "h:\Programming\Coding Practice
\Basic C\exerise class work\3\" ; if ($?) { gcc tempCodeRunnerFile.c -o tempCodeRunnerFile } ; if
($?) { .\tempCodeRunnerFile }
Enter any angle in degree:30
the sin value is: 0.500000
PS H:\Programming\Coding Practice\Basic C\exerise class work\3> |
```



Problem No: 3.3

Problem Name: Write a program that read any angle t and display cos(t).

Source Code:

```
#include<stdio.h>
#include<math.h>
#ifndef M_PI
#define M_PI 3.14159265358979323
#endif
int main() {
    double t,radians;
    printf("Enter any angle in degree:");
    scanf("%lf",&t);
    radians = t * M_PI / 180.0;
    double result=cos(radians);
    printf("the cos value is: %.2lf",result);
    return 0;
}
```



Output:

```
> cd "h:\Programming\Coding Practice
\Basic C\exerise class work\3\" ; if ($?) { gcc code3.c -o code3 } ; if ($?) { .\code3 }
Enter any angle in degree:30
the cos value is: 0.87
PS H:\Programming\Coding Practice\Basic C\exerise class work\3> |
```

Problem No: 3.4

Problem Name: Write a program that read any angle t and display tan(t).

Source Code:

```
#include<stdio.h>
#include<math.h>
int main() {
    double t,pi=3.14159265358979323 ,radians;
    printf("Enter any angle in degree:");
    scanf("%lf",&t);
    radians = t * M_PI /180.0;
    double result=tan(radians);
    printf("the tan value is: %lf",result);
    return 0;
}
```



Output:

```
> cd "h:\Programming\Coding
\Basic C\exercise class work\3\" ; if ($?) { gcc code4.c -o code4 } ; if ($?) { .\code4 }
Enter any angle in degree:45
the tan value is: 1.000000
PS H:\Programming\Coding Practice\Basic C\exercise class work\3> |
```

Problem No: 3.5

Problem Name: Write a program that read any angle t and display cot(t).

Source Code:

```
#include<stdio.h>
#include<math.h>
int main() {
    double t,pi=3.14159265358979323 ,radians;
    printf("Enter any angle in degree:");
    scanf("%lf",&t);
    radians = t * pi /180.0;
    double result=1/tan(radians);
    printf("the cot(%lf) value is:
%.2lf",t,result);
    return 0;
}
```



Output:

```
> cd "h:\Program
\Basic C\exerise class work\3\" ; if ($?) { gcc tempCodeRunnerFile.c -o tempCo
($?) { .\tempCodeRunnerFile }
Enter any angle in degree:30
the cot(30.000000) value is: 1.73
PS H:\Programming\Coding Practice\Basic C\exerise class work\3> |
```

Problem No: 3.6

Problem Name: Write a program that read any angle t and display sec(t).

Source Code:

```
#include<stdio.h>
#include<math.h>
int main() {
    double t,radians,result,pi=3.1416;
    printf("Enter a number in degree: ");
    scanf("%lf",&t);
    radians=t* pi/180.0;
    result= 1/cos(radians);
    printf("the sec(%lf) value is %lf
",t,result);
    return 0;
}
```




Output:

```
> cd "h:\Program
\Basic C\exercise class work\3\" ; if ($?) { gcc tempCodeRunnerFile.c -o tempCo
($?) { .\tempCodeRunnerFile }
Enter a number in degree: 30
the sec(30.000000) value is 1.154701
PS H:\Programming\Coding Practice\Basic C\exercise class work\3> |
```

Problem No: 3.7

Problem Name: Write a program that read any angle t and display cosec(t).

Source Code:

```
#include<stdio.h>
#include<math.h>
int main() {
    double t,radians,result,pi=3.1416;
    printf("Enter a number in degree: ");
    scanf("%lf",&t);
    radians=t* pi/180.0;
    result= 1/sin(radians);
    printf("the cosec(%lf) value is %lf
",t,result);
    return 0;
}
```



Output:

```
> cd "h:\
\Basic C\exercise class work\3\" ; if ($?) { gcc code7.c -o code7 } ; if
Enter a number in degree: 30
the cosec(30.000000) value is 1.999996
PS H:\Programming\Coding Practice\Basic C\exercise class work\3>
```

Problem No: 3.8

Problem Name: Write a program that read a value n and display sin inverse(n).

Source Code:

```
#include<stdio.h>
#include<math.h>
int main() {
    double n,pi=3.14159265358979323;
    printf("Enter any value: ");
    scanf("%lf",&n);
    double result= asin(n)*180.0/pi;
    printf("sin inverse(%lf) = %lf
degree",n,result);
    return 0;
}
```



Output:

```
> cd "h:\Pro
\Basic C\exerise class work\3\" ; if ($?) { gcc tempCodeRunnerFile.c -o te
($?) { .\tempCodeRunnerFile }
Enter any value: 0.5
sin inverse(0.500000) = 30.000000 degree
PS H:\Programming\Coding Practice\Basic C\exerise class work\3> |
```

Problem No: 3.9

Problem Name: Write a program that read a value n and display cos inverse(n).

Source Code:

```
#include<stdio.h>
#include<math.h>
int main(){
    double n,pi=3.14159265358979323;
    printf("Enter any value: ");
    scanf("%lf",&n);
    double result= acos(n)*180.0/pi;
    printf("cos inverse(%.2lf) = %.2lf degree",n,result);
    return 0;
}
```



Output:

```
> cd "h:
\\Basic C\\exerise class work\\3\" ; if ($?) { gcc tempCodeRunnerFile.c -
($?) { .\\tempCodeRunnerFile }
Enter any value: 0
cos inverse(0.00) = 90.00 degree
PS H:\\Programming\\Coding Practice\\Basic C\\exerise class work\\3>
```

Problem No: 3.10

Problem Name: Write a program that read a value n and display tan inverse(n).

Source Code:

```
#include<stdio.h>
#include<math.h>
int main() {
    double n,pi=3.14159265358979323;
    printf("Enter any value: ");
    scanf("%lf",&n);
    double result= atan(n)*180.0/pi;
    printf("tan inverse(%.2lf) = %.2lf
degree",n,result);
    return 0;
}
```



Output:

```
> cd
\\Basic C\\exerise class work\\3\\" ; if ($?) { gcc tempCodeRunnerFile.
($?) { .\\tempCodeRunnerFile }
Enter any value: 1
tan inverse(1.00) = 45.00 degree
PS H:\\Programming\\Coding Practice\\Basic C\\exerise class work\\3>
```

Problem No: 3.11

Problem Name: Write a program that read a value n and display cot inverse(n).

Source Code:

```
#include<stdio.h>
#include<math.h>
int main() {
    double n,pi=3.14159265358979323;
    printf("Enter any value: ");
    scanf("%lf",&n);
    double result= atan(1/n)*180.0/pi;
    printf("cot inverse(%.2lf) = %.2lf
degree",n,result);
    return 0;
}
```



Output:

```
> cd "h:\Prog
\Basic C\exercise class work\3\" ; if ($?) { gcc tempCodeRunnerFile.c -o ter
($?) { .\tempCodeRunnerFile }
Enter any value: 0.5
cot inverse(0.50) = 63.43 degree
PS H:\Programming\Coding Practice\Basic C\exercise class work\3>
```

Problem No: 3.12

Problem Name : Write a program that read a value n and display sec inverse(n).

Source Code:

```
#include<stdio.h>
#include<math.h>
int main(){
    double n,pi=3.14159265358979323;
    printf("Enter any value: ");
    scanf("%lf",&n);
    double result= acos(1.0/n)*180.0/pi;
    printf("sec inverse(%.2lf) = %.2lf
degree",n,result);
    return 0;
}
```



Output:

```
> cd "h:  
\\Basic C\\exercise class work\\3\\" ; if ($?) { gcc code12.c -o code12 } ;  
Enter any value: 6.48  
sec inverse(6.48) = 81.12 degree  
PS H:\\Programming\\Coding Practice\\Basic C\\exercise class work\\3>
```

Problem No: 3.13

Problem Name: Write a program that read a value n and display cosec inverse(n).

Source Code:

```
#include<stdio.h>  
#include<math.h>  
int main() {  
    double n,pi=3.14159265358979323;  
    printf("Enter any value: ");  
    scanf("%lf",&n);  
    double result= asin(1.0/n)*180.0/pi;  
    printf("cosec inverse(%.2lf) = %.2lf  
degree",n,result);  
    return 0;  
}
```



Output:

```
> cd "  
\\Basic C\\exerise class work\\3\" ; if ($?) { gcc code13.c -o code13 }  
Enter any value: 4  
cosec inverse(4.00) = 14.48 degree  
PS H:\\Programming\\Coding Practice\\Basic C\\exerise class work\\3> |
```

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Problem No: 3.14

Problem Name: Write a program that read two numbers(x,y) and display the value of x to the power y.



Source Code:

```
#include<stdio.h>

int main() {

    int x,y;

    printf("Enter base: ");

    scanf ("%d" ,&x) ;

    printf("Enter power: ");

    scanf ("%d" ,&y) ;

    int power=1;

    for(int i=1;i<=y;i++){

        power=power*x;

    }

    printf("%d to the power %d is %d",x,y,power) ;

    return 0;

}
```

Output:

```
> cd "h:\P
\Basic C\exercise class work\3\" ; if ($?) { gcc code14.c -o code14 } ; i
Enter base: 2
Enter power: 8
2 to the power 8 is 256
PS H:\Programming\Coding Practice\Basic C\exercise class work\3> █
```



Problem No: 3.15

Problem Name: Write a program that read any number and display its square root.

Source Code:

```
#include<stdio.h>
#include<math.h>

int main() {
    long n;
    printf("enter any number : ");
    scanf("%ld",&n);
    printf("the square root
is:%ld", (long) sqrt(n));
    return 0;
}
```

Outout:

```
> cd "h:\
\Basic C\exerise class work\3\" ; if ($?) { gcc code15.c -o code15 } ;
enter any number : 9
the square root is:3
PS H:\Programming\Coding Practice\Basic C\exerise class work\3> |
```



Problem No: 3.16

Problem Name: Write a program that read any number x and display e to the power x.

Source Code:

```
#include<stdio.h>
#include<math.h>

int main() {
    float n;
    printf("Enter a number: ");
    scanf("%f", &n);
    printf("the e to the power %f is
    %lf", n, exp(n));
    return 0;
}
```

Output:

```
> cd
\\Basic C\\exerise class work\\3\" ; if ($?) { gcc code16.c -o code16
Enter a number: 2
the e to the power 2.000000 is 7.389056
PS H:\\Programming\\Coding Practice\\Basic C\\exerise class work\\3> 
```



Problem No: 3.17

Problem Name: Write a program that read any number x and display $\log(x)$.

Source Code:

```
#include<stdio.h>
#include<math.h>
int main(){
    float n;
    printf("Enter a number: ");
    scanf("%f",&n);
    printf("log(%f) is %0.2lf",n,log(n));
    return 0;
}
```

Output:

```
> cd
\Basic C\exercise class work\3\" ; if ($?) { gcc tempCodeRunnerFile
($?) { .\tempCodeRunnerFile }
Enter a number: 10
log(10.000000) is 2.30
PS H:\Programming\Coding Practice\Basic C\exercise class work\3>
```



Problem No: 3.18

Problem Name: Write a program that read any number x and display $\log_{10}(x)$.

Source Code:

```
#include<stdio.h>
#include<math.h>

int main() {
    float n;
    printf("Enter a number: ");
    scanf("%f", &n);
    printf("log10(%f) is %0.2lf", n, log10(n));
    return 0;
}
```

Output:

```
> cd
\\Basic C\\exerise class work\\3\\ ; if ($?) { gcc tempCodeRunnerFile
($?) { .\\tempCodeRunnerFile }
Enter a number: 10
log10(10.000000) is 1.00
PS H:\\Programming\\Coding Practice\\Basic C\\exerise class work\\3>
```

if, if...else

Problem No:4.1

Problem Name: Write a program that read an integer and print odd or even.

Source Code:

```
#include<stdio.h>

int main() {
    int n;
    printf("enter a integer value: ");
    scanf("%d",&n);
    if(n%2==0) {
        printf("this number is Even");
    }
    else{
        printf("this number is odd");
    }
    return 0;
}
```



Output:

```
> cd "h
\\Basic C\\exerise class work\\4\\" ; if ($?) { gcc code1.c -o code1 } ;
enter a integer value: 5
this number is odd
PS H:\Programming\Coding Practice\Basic C\\exerise class work\\4> |
```

Problem No: 4.2

Problem Name: Write a program that read two number and display maximum.

Source Code:

```
#include<stdio.h>

int main() {
    int x,y;
    printf("Enter two number: ");
    scanf("%d %d",&x,&y);
    if(x>y) printf("%d is maximum",x);
    else if(x<y) printf("%d is maximum",y);
    else if(x=y) printf("the number is equal");
    return 0;
}
```



Output:

```
> cd "h
\Basic C\exercise class work\4\" ; if ($?) { gcc code2.c -o code2 } ;
Enter two number: 100
900
900 is maximum
PS H:\Programming\Coding Practice\Basic C\exercise class work\4> |
```

Problem No: 4.3

Problem Name: Write a program that read two number and display maximum.

Source Code:

```
#include<stdio.h>

int main() {
    int a,b;
    printf("Enter first number: ");
    scanf("%d",&a);
    printf("Enter second number: ");
    scanf("%d",&b);
    if(a>b) printf("First number %d is maximum",a);
    else if(a<b) printf("Second number %d is
maximum",b);
    else if(a==b) printf("the number is equal");
    return 0; }
```




Output:

```
> cd "h:
\\Basic C\\exerise class work\\4\\" ; if ($?) { gcc code3.c -o code3 } ; i
Enter first number: 900
Enter second number: 100000
Second number 100000 is maximum
PS H:\Programming\Coding Practice\Basic C\\exerise class work\\4> |
```

Problem No: 4.4

Problem Name: Write a program that read two number and display minimum.

Source Code:

```
#include<stdio.h>

int main() {
    int a,b;
    printf("Enter first number: ");
    scanf("%d",&a);
    printf("Enter second number: ");
    scanf("%d",&b);
    if(a<b) printf("First number %d is minimum",a);
    else if(a>b) printf("Second number %d is
minimum",b);
    else if(a=b) printf("the number is equal");
    return 0;}
```



Output:

```
> cd
\Basic C\exercise class work\4\" ; if ($?) { gcc code4.c -o code4 }
Enter first number: 2
Enter second number: 10
First number 2 is minimum
PS H:\Programming\Coding Practice\Basic C\exercise class work\4>
```

Problem No: 4.5

Problem Name: Write a program that read two number and display minimum.

Source Code:

```
#include<stdio.h>

int main() {
    double a,b;
    printf("Enter first number: ");
    scanf("%lf",&a);
    printf("Enter second number: ");
    scanf("%lf",&b);

    if(a<b) printf("First number %.2lf is minimum",a);
    else if(a>b) printf("Second number %.2lf is minimum",b);
    else if(a==b) printf("the number is equal");

    return 0;}

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```

Output:

```
> cd "h
\Basic C\exercise class work\4\" ; if ($?) { gcc code5.c -o code5 } ;
Enter first number: 455
Enter second number: 655
First number 455.00 is minimum
PS H:\Programming\Coding Practice\Basic C\exercise class work\4>
```

Problem No:4.6

Problem Name: Write a program that read three number and display maximum.

Source Code:

```
#include<stdio.h>
int main() {
    int x,y,z;
    printf("Enter first number: ");
    scanf("%d",&x);
    printf("Enter second number: ");
    scanf("%d",&y);
    printf("Enter third number: ");
    scanf("%d",&z);
```

```

    if(x>y) {
        if(x>z) printf("first number %d is
maximum",x) ;
        else printf("third number %d is
maximum",z) ;
    }
    else{
        if(y>z) printf("second number %d is
maximum",y) ;
        else printf("third number %d is
maximum",z) ;
    }return 0;
}

```

Output:

```

> cd
\Basic C\exercise class work\4\" ; if ($?) { gcc code6.c -o code6 }
Enter first number: 22
Enter second number: 66
Enter third number: 44
second number 66 is maximum
PS H:\Programming\Coding Practice\Basic C\exercise class work\4>

```



Problem No:4.7

Problem Name: Write a program that read three number and display minimum.

Source Code:

```
#include<stdio.h>

int main() {

    int x,y,z;

    printf("Enter first number: ");
    scanf("%d",&x);

    printf("Enter second number: ");
    scanf("%d",&y);

    printf("Enter third number: ");
    scanf("%d",&z);

    if(x<y) {

        if(x<z) printf("first number %d is
minimum",x);

        else printf("third number %d is
minimum",z);

    }

    else{
```

```

        if(y<z) printf("second number %d is
minimum",y);

        else printf("third number %d is
minimum",z);

    }return 0;

}

```

Output:

```

> cd "h:
\Basic C\exerise class work\4\" ; if ($?) { gcc code7.c -o code7 } ; i
Enter first number: 66
Enter second number: 44
Enter third number: 22
third number 22 is minimum
PS H:\Programming\Coding Practice\Basic C\exerise class work\4>

```

Problem No:4.8

Problem Name: Write a program that read three number and display medium.

Source Code:

```

#include<stdio.h>

int main(){

    int x,y,z;

    printf("Enter first number: ");

    scanf("%d",&x);

    printf("Enter second number: ");

    scanf("%d",&y);

```

```

printf("Enter third number: ");
scanf("%d",&z);
if(x>y){
    if(x>z){
        if(y>z) printf("midium number is %d",y); //
z<y<x
        else printf("midium number is %d",z); //
y<z<x
    }
    else printf("midium number is %d",x); // y<x<z
}
else{ // y>x
    if(y>z){
        if(x>z) printf("midium number is %d",x);//
y>x>z ,
        else printf("midium number is %d",z);//
y>z>x
    }
    else printf("midium number is %d",y); // z>y>x
}
return 0;
}

```



Output:

```
> cd "h:\P
\Basic C\exercise class work\4\" ; if ($?) { gcc code8.c -o code8 } ; if
Enter first number: 22
Enter second number: 66
Enter third number: 44
midium number is 44
PS H:\Programming\Coding Practice\Basic C\exercise class work\4>
```

Problem No:4.9

Problem Name: Write a program that read mark and display pass or fail.

Source Code:

```
#include<stdio.h>

int main() {
    int n;

    printf("enter any number you get in exam:");
    scanf("%d",&n);

    if(n>=40) printf("you are pass");
    else if(n<40) printf("you are fail");
    return 0;
}
```




Output:

```
> cd "h
\Basic C\exercise class work\4\" ; if ($?) { gcc code9.c -o code9 } ;
enter any number you get in exam:80
you are pass
PS H:\Programming\Coding Practice\Basic C\exercise class work\4> |
```

Problem No: 4.10

Problem Name: Write a program that read mark and display result in devision.

Source Code:

```
#include<stdio.h>

int main() {
    int n;
    printf("Enter any mark: ");
    scanf("%d", &n);
    if(n>=80) printf("First Division");
    else if(n>=60) printf("Second Division");
    else if(n>=40) printf("Third Division");
    else printf("Fail");
    return 0;}
```



Output:

```
> cd "h:\
\Basic C\exercise class work\4\" ; if ($?) { gcc code10.c -o code10 } ;
Enter any mark: 80
First Division
PS H:\Programming\Coding Practice\Basic C\exercise class work\4>
```

Problem No: 4.11

Problem Name: Write a program that read mark and display result in grade.

Source Code:

```
#include<stdio.h>

int main() {
    int n;

    printf("Enter any mark: ");
    scanf("%d",&n);

    if(n>=80) printf(" A Grade");
    else if(n>=70) printf("B Grade");
    else if(n>=60) printf("C Grade");
    else if(n>=40) printf("D Grade");
    else printf("Fail");

    return 0;}

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```



Output:

```
> cd "h:\
\Basic C\exerise class work\4\" ; if ($?) { gcc code11.c -o code11 } ;
Enter any mark: 80
A Grade
PS H:\Programming\Coding Practice\Basic C\exerise class work\4>
```

Problem No: 4.12

Problem Name: Write a program that read any year and display leap year or not.

Source Code:

```
#include<stdio.h>

int main() {
    int n;

    printf("Enter any year: ");
    scanf("%d", &n);

    if( (n%4==0) || (n%100==0) || (n%400==0) )
printf("the year is leap year");

    else printf("the year is not leap year");

    return 0;
}
```



Output:

```
> cd
\\Basic C\\exercise class work\\4\" ; if ($?) { gcc code12.c -o code12
Enter any year: 2025
the year is not leap year
PS H:\\Programming\\Coding Practice\\Basic C\\exercise class work\\4> |
```

Problem No: 4.13

Problem Name: Write a program that read three number(a,b,c) and determine the roots of the quadratic equation: $ax^2+bx+c=0$

Source Code:

```
#include<stdio.h>
#include<math.h>

int main() {
    int a,b,c;

    printf("Enter a: ");
    scanf("%d",&a);
    printf("Enter b: ");
    scanf("%d",&b);
    printf("Enter c: ");
```

```
scanf ("%d", &c) ;

if (a==0) {

    printf(" a can't be zero in quadratic
equation");

    return 1;

}

double d,r1,r2,x,y; //
discriminent,root1,root2,real_part,imaginary_par
t;

d= b*b-4*a*c;

if (d>0) {

    r1=(-b+sqrt(d))/2*a;
    r2=(-b-sqrt(d))/2*a;

    printf("roots are real and differnt");
    printf("root 1 is %.11f\n",r1);
    printf("root 2 is %.11f",r2);

}

else if (d==0){

    r1=-b/2*a;

    printf("Both root are same");

    printf("%.11f",r1);

}

else{
```

```

        x=-b/2*a;
        y=sqrt(d)/2*a;
        printf("roots are complex root\n");
        printf("r1=%.11f+%.11fi \n r2=%.11f-
%.11fi",x,y,x,y);
    }
    return 0;
}

```

Output:

```

> cd "h:\P
\Basic C\exercise class work\4\" ; if ($?) { gcc code13.c -o code13 } ; i
Enter a: 2
Enter b: 6
Enter c: 4
roots are real and differentroot 1 is -4.0
root 2 is -8.0
PS H:\Programming\Coding Practice\Basic C\exercise class work\4>

```

Conditional Operator

Problem No: 5.1

Problem Name: Write a program that read an integer and print odd or even.

Source Code:

```
#include<stdio.h>

int main() {
    int n;
    printf("Enter any integer number: ");
    scanf("%d",&n);
    n%2==0? printf("even number"):printf("odd
number");
    return 0;
}
```

Output:

```
PS H:\Programming\Coding Practice\Basic C> cd "h:\Programming\Coding Practice\class work\5\" ; if ($?) { gcc code1.c -o code1 } ; if ($?) { .\code1 }
Enter any integer number: 76
even number
PS H:\Programming\Coding Practice\Basic C\exercise class work\5> █
```



Problem No: 5.2

Problem Name: Write a program that read two number and display maximum.

Source Code:

```
#include<stdio.h>

int main() {

    int a,b,max;

    printf("Enter two numbers: ");

    scanf("%d %d",&a,&b);

    a>b? printf("%d is max",a):printf("%d is max",b);

    return 0;

}
```

Output:

```
> cd "h:\
\Basic C\exercise class work\5\" ; if ($?) { gcc code2.c -o code2 } ; if
Enter two numbers: 66
44
66 is max
PS H:\Programming\Coding Practice\Basic C\exercise class work\5>
```




Problem No: 5.3

Problem Name: Write a program that read two number and display maximum.

Source Code:

```
#include<stdio.h>

int main() {

    double a,b,max;

    printf("Enter two numbers: ");

    scanf("%lf %lf",&a,&b);

    max=(a>b) ? a:b;

    printf("max=%.2lf",max);

    return 0;

}
```

Output:

```
> cd "h:\P
\Basic C\exercise class work\5\" ; if ($?) { gcc code3.c -o code3 } ; if
Enter two numbers: 10
100
max=100.00
PS H:\Programming\Coding Practice\Basic C\exercise class work\5>
```



Problem No: 5.4

Problem Name: Write a program that read two number and display minimum.

Source Code:

```
#include<stdio.h>

int main() {

    double x,y;

    printf("Enter two different number:");

    scanf("%lf %lf",&x,&y);

    double min=(x<y) ? x : y ;

    printf("%.2lf is minimun",min);

    return 0;

}
```

Output:

```
> cd "h:\F
\Basic C\exerise class work\5\" ; if ($?) { gcc code4.c -o code4 } ; if
Enter two different number:9
10
9.00 is minimun
PS H:\Programming\Coding Practice\Basic C\exerise class work\5> |
```



Problem No: 5.5

Problem Name: Write a program that read two number and display minimum.

Source Code:

```
#include<stdio.h>

int main() {

    int x,y;

    printf("Enter two different number:");

    scanf("%d %d",&x,&y);

    int min=(x<y) ? x : y;

    printf("%d is minimun",min);

    return 0;

}
```

Output:

```
> cd "
\\Basic C\\exerise class work\\5\\" ; if ($?) { gcc code5.c -o code5 } ;
Enter two different number:88
99
88 is minimun
PS H:\Programming\Coding Practice\Basic C\exerise class work\5>
```



Problem No: 5.6

Problem Name: Write a program that read three number and display maximum.

Source Code:

```
#include<stdio.h>

int main() {

    int a,b,c,max;

    printf("enter three different number: ");

    scanf("%d%d%d",&a,&b,&c);

    max=(a>b)? (a>c)? a:b: (b>c)? b:c;

    printf("%d is maximum",max);

    return 0;

}
```

Output:

```
> cd "h
\Basic C\exerise class work\5\" ; if ($?) { gcc code6.c -o code6 } ;
enter three different number: 9
8
7
9 is maximum
PS H:\Programming\Coding Practice\Basic C\exerise class work\5> |
```



Problem No: 5.7

Problem Name: Write a program that read three number and display minimum.

Source Code:

```
#include<stdio.h>

void main() {

    int a,b,c,min;

    printf("enter three different number: ");

    scanf("%d%d%d",&a,&b,&c);

    min=(a<b)? (a<c)? a:b:(b<c)? b:c;

    printf("%d is minimum",min);

}
```

Output:

```
> cd "h
\\Basic C\\exerise class work\\5\" ; if ($?) { gcc code7.c -o code7 } ;
enter three different number: 5
7
9
5 is minimum
PS H:\Programming\Coding Practice\Basic C\exerise class work\5> |
```



Problem No: 5.8

Problem Name: Write a program that read three number and display medium.

Source Code:

```
#include<stdio.h>

int main() {

    int a,b,c,mid;

    printf("enter three differnt number: ");

    scanf("%d%d%d",&a,&b,&c);

    mid=(a>b)? (a>c)? (b>c)? b:c:a: (b>c)?
(a>c)? a:c:b;

    printf("%d is medium",mid);

    return 0;

}
```

Output:

```
> cd "h:\
\Basic C\exerise class work\5\" ; if ($?) { gcc code8.c -o code8 } ; if
enter three differnt number: 99
100
12
99 is medium
PS H:\Programming\Coding Practice\Basic C\exerise class work\5> |
```



Problem No: 5.9

Problem Name: Write a program that read mark and display pass or fail.

Source Code:

```
#include<stdio.h>

int main() {

    int mark;

    printf("enter a number you got: ");

    scanf("%d",&mark);

    (mark>=33)? printf("Pass"):printf("Fail");

    return 0;

}
```

Output:

```
> cd "h:\
\Basic C\exercise class work\5\" ; if ($?) { gcc code9.c -o code9 } ; if
enter a number you got: 40
Pass
PS H:\Programming\Coding Practice\Basic C\exercise class work\5> |
```

Thank You